

Name: Calvin, Andrew, Dylan

Date: 5/13/2024

Class: Engineering Period 3

EDPPSR: Element I

Problem Statement

The Walpole High School Film Crew need a better way to protect the cameras while they are not in use. Currently, the cameras are susceptible to damage from factors like dust particles or scratches, which can cause damage to the lens. This can negatively impact the picture quality, as videos produced on a scratched or dusty camera end up looking less clear than if the lens was intact. For this reason, we designed a cover to shield the camera from factors such as this and keep the lens safe.

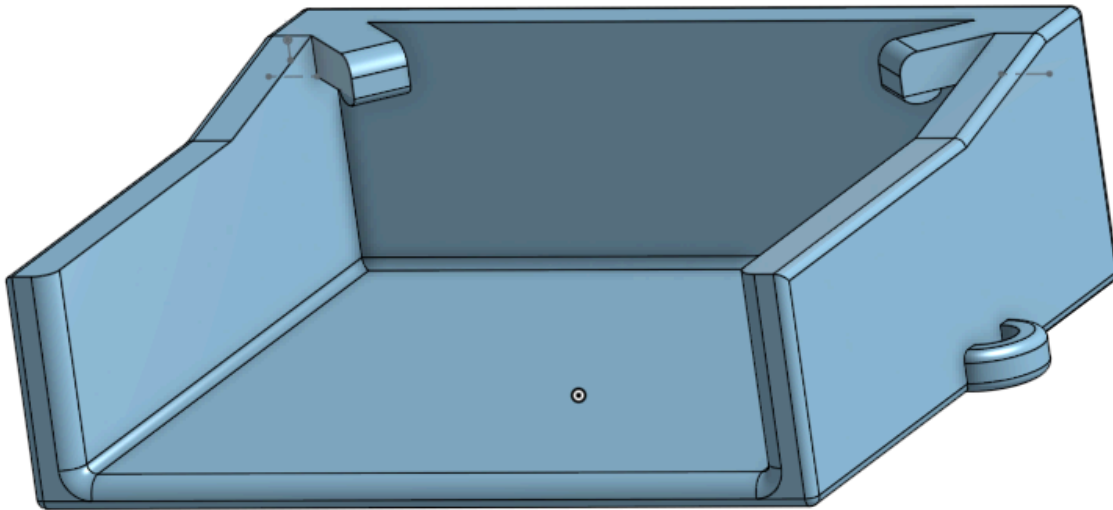
Prototype 1

Our first prototype that we did testing on is the one with the bigger walls. We created a model of the design on OnShape, and printed out our design with an Afinia 3D printer. Andrew tested this design by throwing it in the air, and testing if the impact from the fall did any damage to the walls of the cover. It was able to withstand the first couple of falls easily, but by the time it reached 8'5.5", the walls had started to chip. However, the damage was not nearly enough to penetrate the wall completely.

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Because of this, we decided to make the walls slightly thicker. We altered the settings in Prusa, making it so the walls were more durable.

This new version of the design survived the durability test where the previous one did not, meaning that it was more durable and passed the drop test.

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	requirement	testing setup and procedure	test cases and number of trials	expected range of results	pass/fail criteria, goal	Testing Results 1	Testing Results 2	Testing Results 3
High Priority 1	Protect the camera	Put the cover on the camera and see if the lens gets damaged with it on	3	Camera is fully protected, camera still has damage on it but less now, camera has same amount of damage on it as before	Lens remains undamaged	Pass - no damage done to the camera lens, lens cover doesn't let anything get to lens	pass	pass
2	Stay on the lens when on	Shake the camera when the cover is on, see if the cover falls off or not	3	Cover stays on, cover falls off	No slippage off of the lens	Pass - shook it thoroughly and the lens cover did not fall off	pass	pass
3	Stay attached to the camera when not attached to lens	Move the camera around a lot and see if the cover falls off or not	1	Cover stays on, cover falls off	The cover doesn't detach from the camera	Pass - The lens is attached to the camera by a string with a band that goes around the camera		
4	Easily removable and attachable	Attempt to attach and remove it	5	Cover is easily attachable and removable, cover is attachable and	The cover doesn't stick to the lens for more than 0.5 seconds	Pass - absolutely no difficulty removing and replacing the	pass	pass

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				removable with some difficulty, cover can't be attached or removed from the camera		lens cover		
5	Durability	Attempt to damage it in ways that would occur naturally to a user of the product	5	Cover has no scratches on it, cover is mildly damaged, cover is very fragile	There are no major chips or cracks in the cover	Fail - cracks appeared and outer wall started to peel off	pass	pass
6	Color	Have the camera be black on the inside, while being a more bright and visible color on the outside	2	Cover is easily visible on outside and completely black on inside, cover is somewhat bright on outside or somewhat black on inside, cover is not visible at all on outside or not black at all on inside.	The cover is a solid black in color	Fail - cover was black on inside, but we didn't get the idea to make the outside bright until after Design-A-Thon feedback.		
7	Low weight	Put on scale	1	<50 grams, <100 grams, >100 grams	The lens cover weighs less than 50 grams	Pass, under 50 grams		

1 Durability Drop Test

High Priority

We dropped the product from the ceiling of the room, which is about 10 feet tall 5 times, and then we inspected the product for any damage after the tests.

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For the first time that we ran the test, the 3d printed material cracked and a piece of the outer shell broke off.



This is a picture of the damage. The next step for our design process was to double the walls on the 3d print. This makes the inner and outer walls that surround the infill thicker so that it is harder for them to break.

After we made this change to the 3d print, we did the same test, and the product did not break or crack at all.

	Damage from 10 foot drop
Prototype 1	yes
Prototype 2	no

Shake Test 1

The next test that we did was the shake test. This was to check if the clips that held the cover onto the back of the box around the lens were secure. This was a very high priority test, as you can't have a functioning lens cover if it just keeps falling off. The test included violently shaking the lens cover when attached to the camera for about 10 seconds. The camera lens cover never detached or slipped, and passed all tests.